

Chapter 3

Charts Depicting Data Visually

Summary of problem:

- In this exercise you will learn how to work with charts.
- You will have computer-related jobs data and you need to analyze them. This data is published by the Bureau of Labor Statistics on the U.S. Department of Labor's website.
- This dataset includes seven jobs, the number of those jobs in 2010, the projected number of jobs by 2020, the growth in percentage increase and number of jobs, and the 2010 median pay.
- You will create several charts that depict the job growth in the computer industry. You will see that different charts provide different perspectives on the data.

Step 1: CREATE A CLUSTERED COLUMN CHART

You want to compare the number of jobs in 2010 to the projected number of jobs in 2020 for all seven computer-related professions entered into the worksheet. You decide to create a clustered column chart to depict this data. After you create this chart, you will move it to its own chart sheet.

a. Open e03h1Jobs

b. Select the range A5:D12.

You selected the job titles, the number of jobs in 2010, the projected number of jobs in 2020, and the number of new jobs. Because you are selecting three data series (three columns of numerical data), you must also select the column headings on row 5.

c. Click Quick Analysis at the bottom-right corner of the selected range and click Charts. The Quick Analysis gallery displays recommended charts based on the selected range.

d. Point to Clustered Column (the third thumbnail in the Charts gallery) to see a preview of what the chart would look like and click Clustered Column. Excel inserts a clustered column chart based on the selected data. The Design tab displays on the Ribbon while the chart is selected.

e. Click Move Chart in the Location group.

The Move Chart dialog box opens for you to specify where to move the chart.

f. Click New sheet, type Column Chart, and then click OK. Save the workbook. Excel moves the clustered column chart to a new sheet called Column Chart.

The screenshot shows the Excel interface with the 'e03h1Jobs' worksheet. The data table is as follows:

	# of Jobs		Job Growth		Median Pay
	2010	2020 Est.	# of New Jobs	% Growth	2010
Systems Analysts	544,400	664,500	120,400	22%	\$77,380
Software App Developers	520,800	664,500	143,700	28%	
Programmers	363,200	406,800	43,600	12%	
Network/System Admins	347,200	443,800	96,600	28%	
CIS Managers	307,900	343,700	35,800	12%	
Info Security Analysts	302,300	347,900	45,600	15%	
Database Administrators	110,800	144,800	34,000	31%	

Annotations in the image:

- Step b:** Select the range (points to the data table).
- Step d:** Design tab (points to the 'Design' tab on the ribbon).
- Step d:** Clustered column chart (points to the chart thumbnail in the Quick Analysis gallery).
- Step e:** Click to move chart (points to the 'Move Chart' button in the 'Location' group).
- Step f:** Type sheet name (points to the 'New sheet' option in the 'Move Chart' dialog box).

The 'Move Chart' dialog box is open, showing the 'New sheet' option selected, with the sheet name 'Column Chart' entered.

STEP 2: CREATE A BAR CHART

You want to create a bar chart to depict the number of jobs in 2010 and the number of new jobs that will be created by 2020. Finally, you want to change the chart to a stacked bar chart to show the total jobs in 2020 based on the number of jobs in 2010 and the number of new jobs.

a. Click the Outlook sheet tab, select the range A5:B12, press and hold Ctrl, and then select the range D5:D12. You used Ctrl to select nonadjacent ranges: the job title labels, the number of jobs in 2010, and the number of new jobs.

TIP: PARALLEL RANGES Nonadjacent ranges should be parallel so that the legend will correctly reflect the data series. This means that each range should contain the same number of related cells. For example, A5:A12, B5:B12, and D5:D12 are parallel ranges. Even though cell A5 is blank, you must select it to have a parallel range with the other two selected ranges that include cells on row 5.

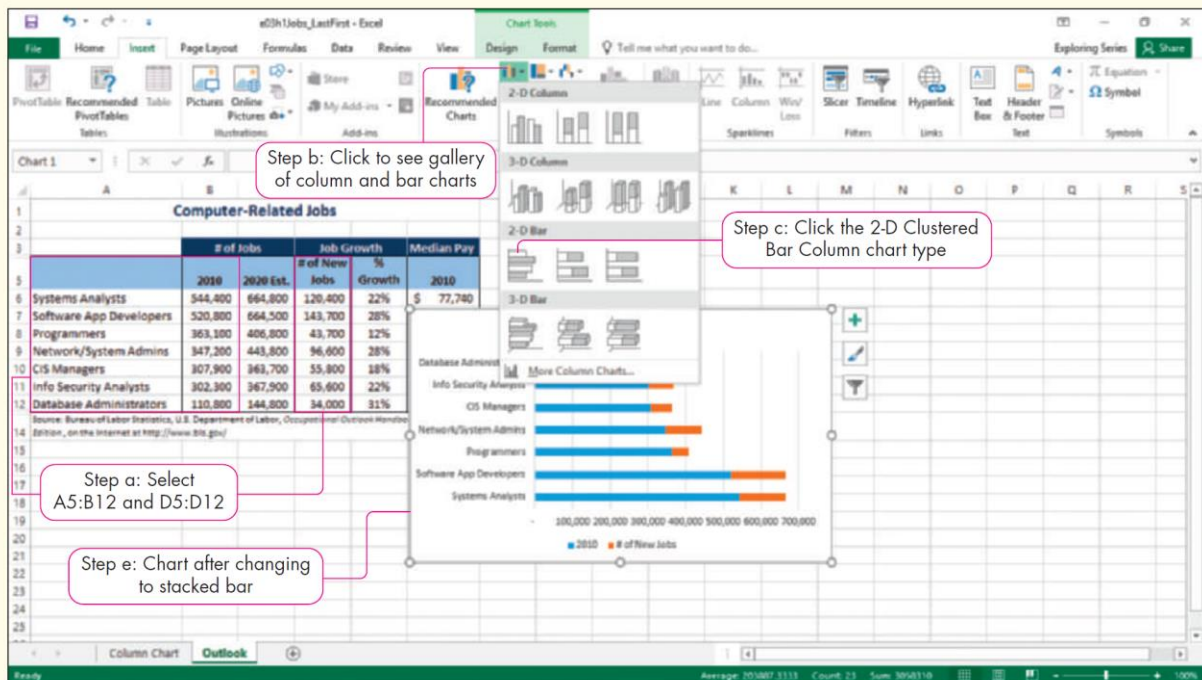
b. Click the Insert tab and click Insert Column or Bar Chart in the Charts group.

c. Click Clustered Bar in the 2-D Bar section to create a clustered bar chart. Excel inserts the clustered bar chart in the worksheet.

d. Click Change Chart Type in the Type group on the Design tab.

The Change Chart Type dialog box opens. The left side of the dialog box lists all chart types. The top-right side displays thumbnails of various bar charts, and the lower section displays a sample of the selected chart.

e. Click Stacked Bar in the top center of the dialog box and click OK. Save the workbook. Excel displays the number of jobs in 2010 in blue and stacks the number of new jobs in orange into one bar per job title. This chart tells the story of how the total projected number of jobs in 2020 is calculated: the number of existing jobs in 2010 (blue) and the number of new jobs (orange).

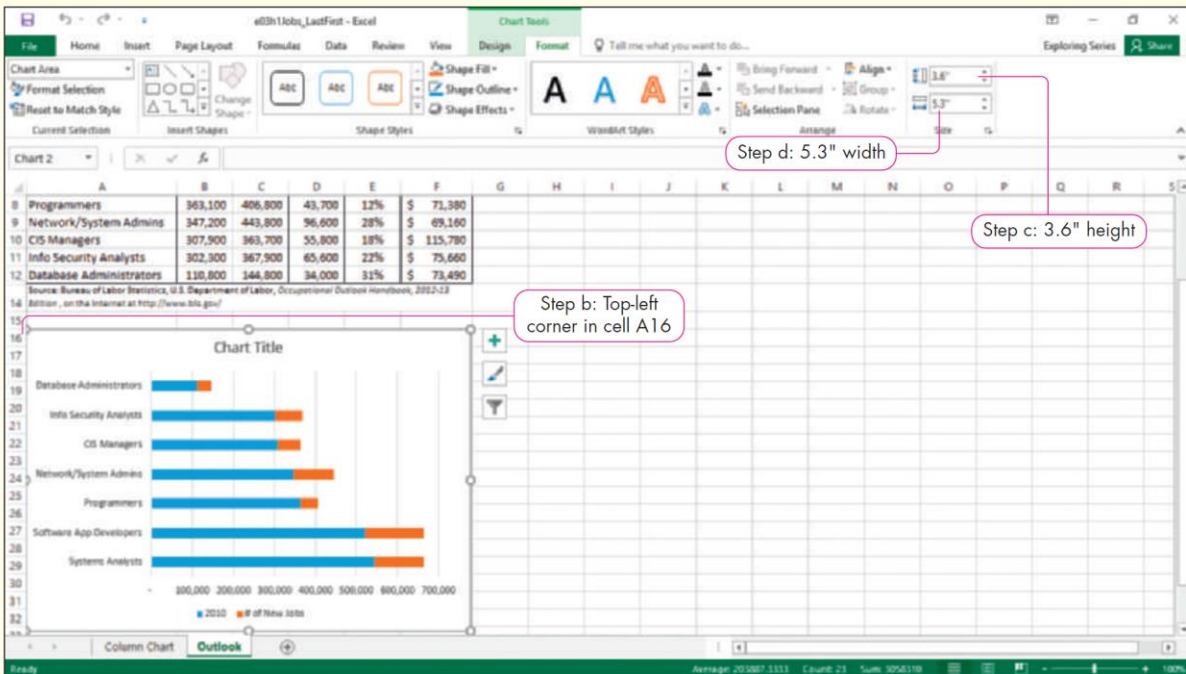


Excel 2016, Windows 10, Microsoft Corporation

STEP 3: MOVE AND SIZE A CHART

The bar chart is displayed in the middle of the worksheet. You decide to position it below the job outlook data and adjust its size to make it larger so that it is as wide as the dataset and a little taller for better proportions.

- Point to an empty part of the chart area. The pointer displays a four-headed arrow with the regular white arrowhead, and the Chart Area ScreenTip displays.
- Drag the chart so that the top-left corner of the chart appears in cell A16. You positioned the chart below the worksheet data.
- Click the Format tab, select the value in the Shape Height box, type 3.6, and then press Enter. The chart is now 3.6" tall.
- Select the value in the Shape Width box, type 5.3, and then press Enter. Save the workbook. The chart is now 5.3" wide.



Excel 2016, Windows 10, Microsoft Corporation

STEP 4: CREATE A PIE CHART

You decide to create a pie chart that depicts the percentage of new jobs by job title calculated from the total number of new jobs created for the seven job titles. After creating the pie chart, you will move it to its own sheet.

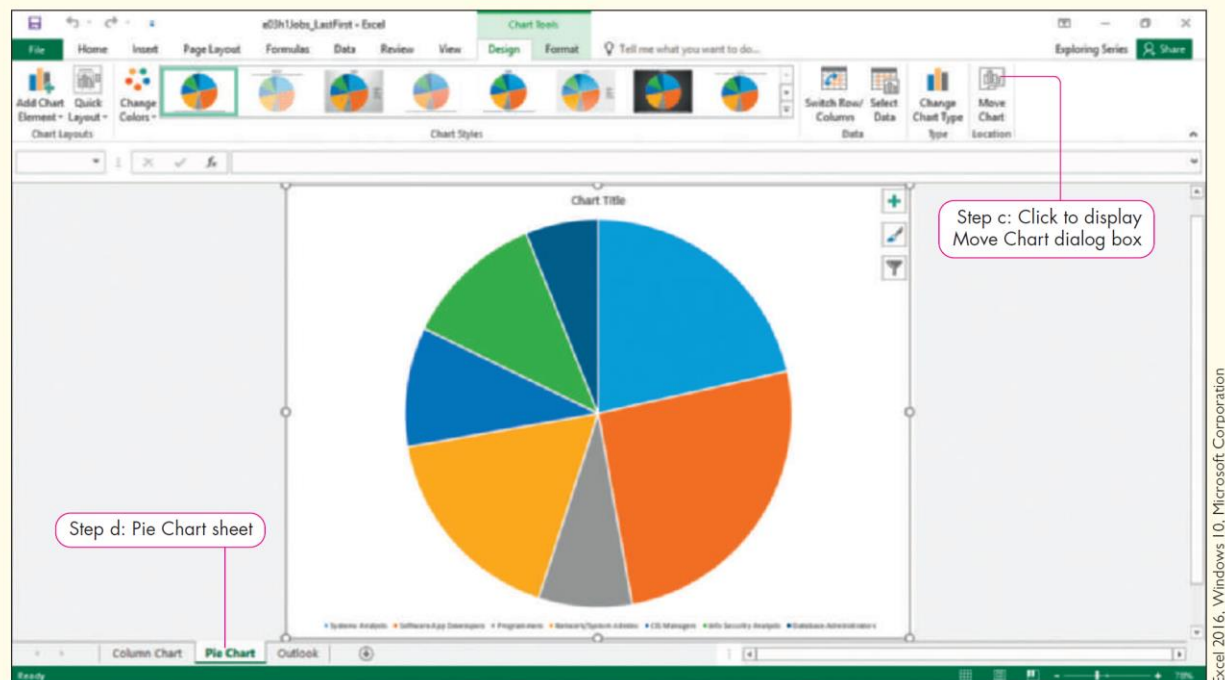
a. Select the range A6:A12, press and hold Ctrl, then select the range D6:D12.

TROUBLESHOOTING: Do not select cells A5 and D5 this time because you are creating a pie chart. When creating a chart from a single data series (e.g., # of New Jobs), you do not need to select the column headings.

b. Click the Insert tab, click Insert Pie or Doughnut Chart in the Charts group, and then select Pie in the 2-D Pie group on the gallery. The pie chart displays in the worksheet.

c. Click Move Chart in the Location group on the Design tab. The Move Chart dialog box opens.

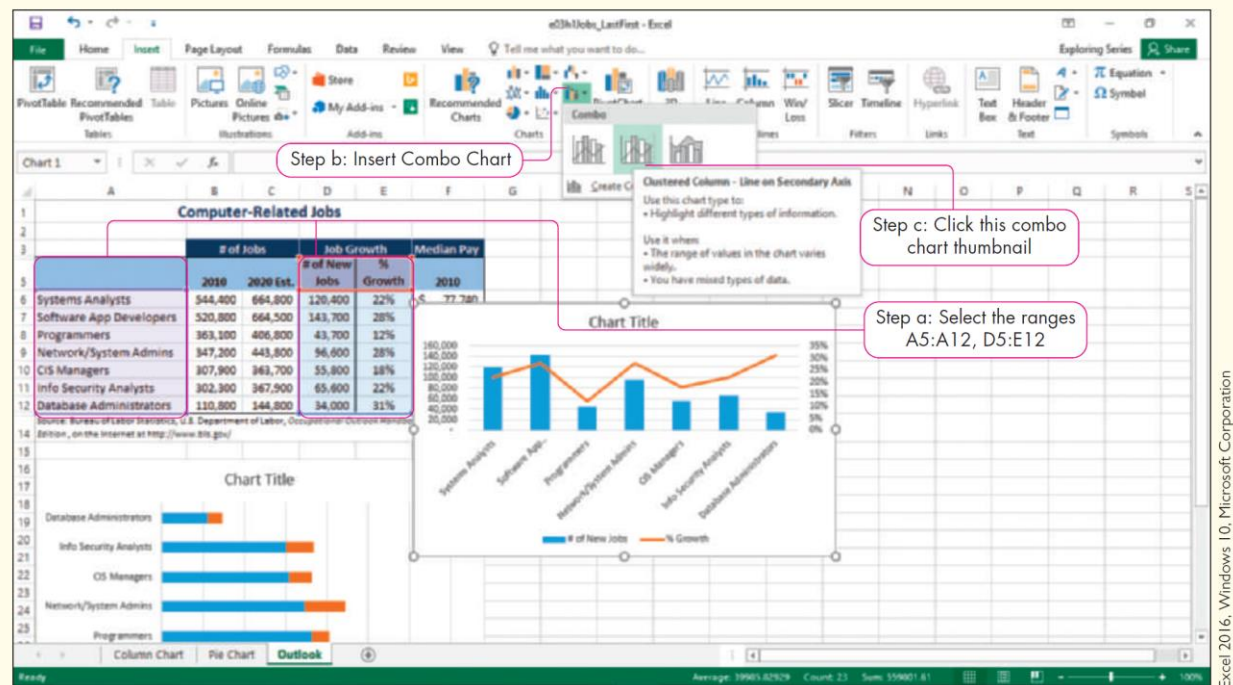
d. Click New sheet, type Pie Chart, and then click OK. Save the workbook. Excel creates a new sheet called Pie Chart.



STEP 5: CREATE A COMBO CHART

You want to create a combo chart that shows the number of new jobs in columns and the percentage of new jobs created in a line on the secondary axis. Although the number of new jobs may appear low as represented by the smallest column (such as 34,000 new database administrators), the actual percentage of new jobs created between 2010 and 2020 may be significant as represented by the steep incline of the orange line (such as 31% growth for database administrators).

- a.** Click the Outlook sheet tab, select the range A5:A12, press and hold Ctrl, then select the range D5:E12.
- b.** Click the Insert tab and click Insert Combo Chart in the Charts group. The Combo Chart gallery of thumbnails displays.
- c.** Click the Clustered Column – Line on Secondary Axis thumbnail, which is the middle thumbnail. Excel creates a combo chart based on the thumbnail you selected. The number of new jobs displays in blue columns, and the percentage growth displays as an orange line.
- d.** Click Move Chart in the Location group on the Design tab, click New sheet, type Combo Chart, and then click OK.
- e.** Save the workbook.



Step 6: EDIT AND FORMAT CHART TITLES

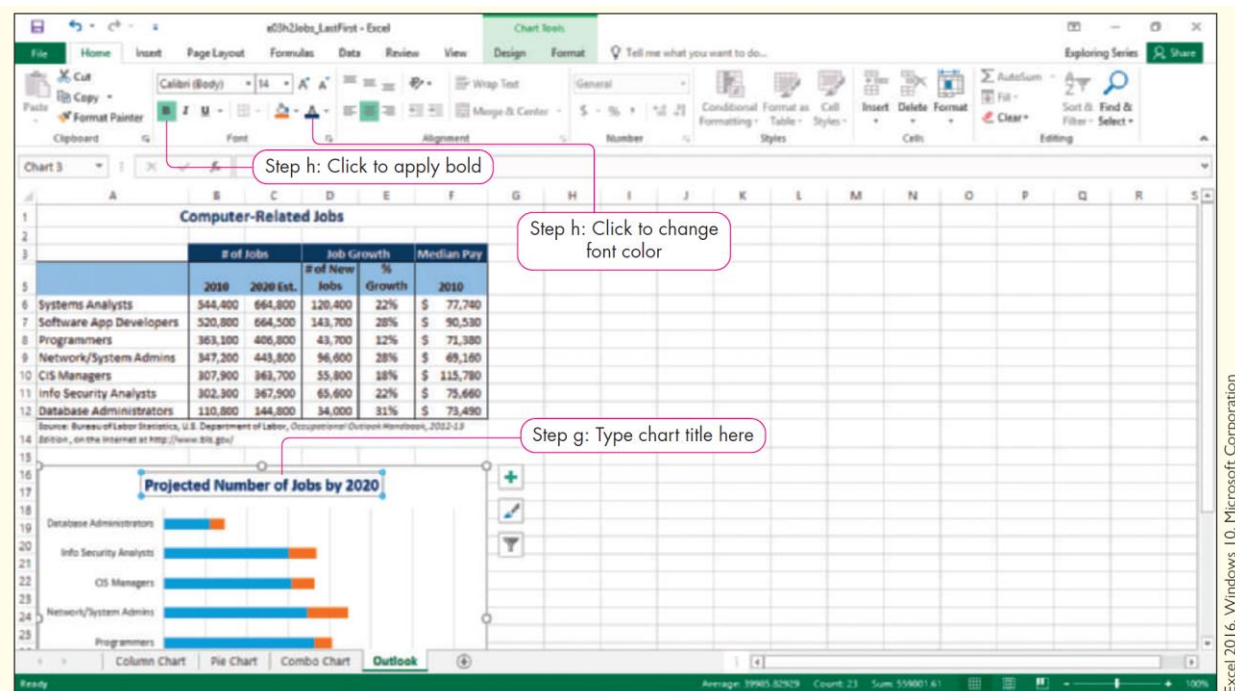
When you created the column, bar, and pie charts in Hands-On Exercise 1, Excel displayed Chart Title at the top of each chart. You will add a title that appropriately describes each chart. In addition, you want to format the chart titles by applying bold and enlarging the font sizes.

- Make sure the **Combo Chart** sheet is the active sheet, select the Chart Title placeholder, type Number of New Computer-Related Jobs by 2020, and then press Enter. As you type a chart title, Excel displays the text in the Formula Bar. The text does not appear in the chart title until after you press Enter.
- Click the Home tab, click Bold, click the Font Color arrow, and then select Black, Text 1. You applied font formats so that the chart title stands out.
- Click the **Pie Chart** sheet tab, select the Chart Title placeholder, type New Computer Related Jobs by 2020, and then press Enter. Excel displays the text you typed for the chart title.
- Click the Home tab, click Bold, click the Font Size arrow and select 18, and then click the Font Color arrow and select Black, Text 1. You formatted the pie chart title so that it stands out.

e. Click the **Column Chart** sheet tab, select the Chart Title placeholder, type Number of Computer-Related Jobs 2010 and 2020, and then press Enter. Click Bold, click the Font Size arrow, and then select 18. Click the Font Color arrow and click Black, Text 1 font color to the chart title.

f. Click the **Outlook** sheet tab, select the Chart Title placeholder, type Projected Number of Jobs by 2020, and then press Enter.

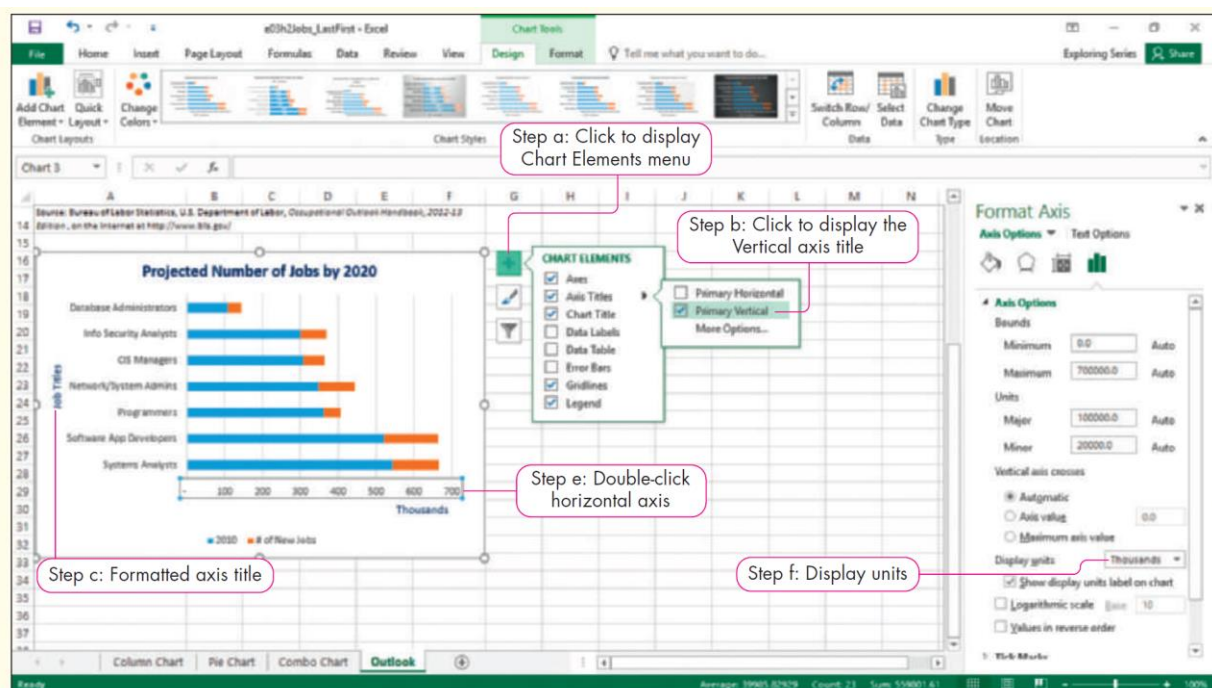
g. Click Bold, click the Font Size arrow, and then select 14. Click the Font Color arrow and click Dark Blue in the Standard Colors section. Save the workbook.



STEP 7: ADD AND FORMAT AXIS TITLES AND FORMAT AXES

For the bar chart, you want to add and format a title to describe the job titles on the vertical axis. In addition, you want to simplify the horizontal axis values to avoid displaying, 000 for each increment and add the title Thousands.

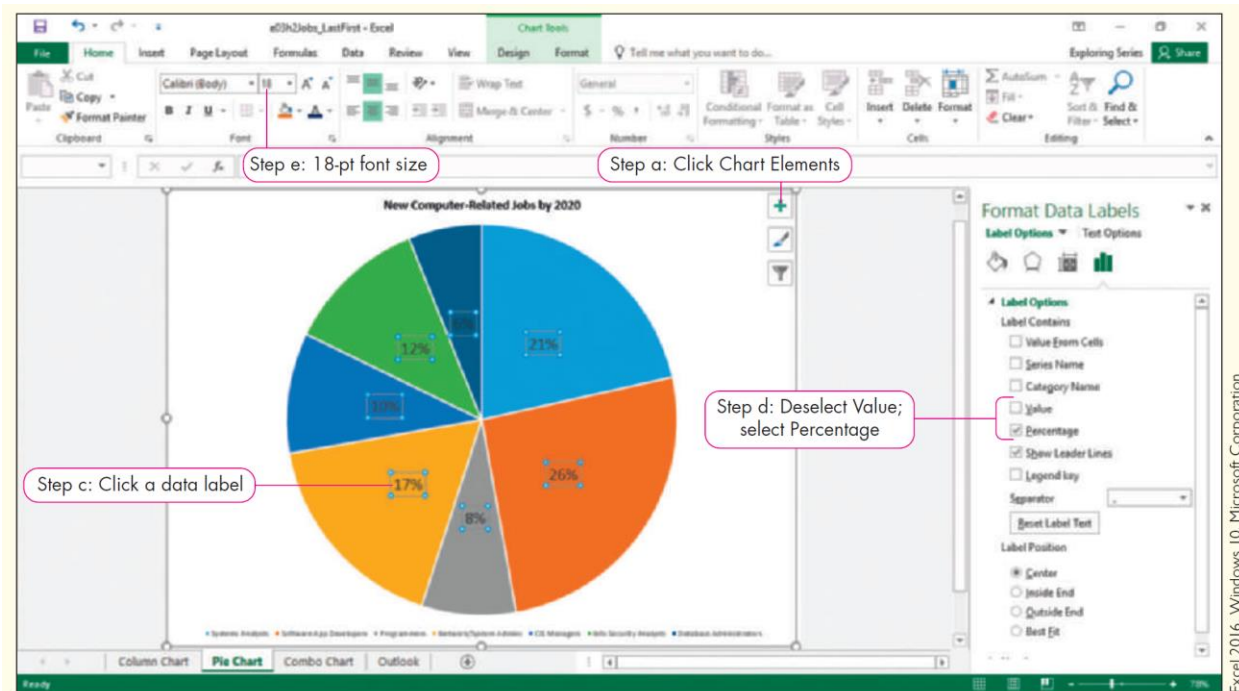
- Ensure that the bar chart is selected in the **Outlook** worksheet and click Chart Elements to the right of the chart. Excel displays the Chart Elements menu.
- Point to Axis Titles, click the Axis Titles arrow, and then click the Primary Vertical check box to select it. Close the menu. Excel displays Axis Title on the left side of the vertical axis.
- Ensure that the Axis Title placeholder is selected, type Job Titles, and then press Enter.
- Click Font Color to apply the default Dark Blue font color to the selected axis title.
- Point to the horizontal axis. When you see the ScreenTip, Horizontal (Value) Axis, double-click the values on the horizontal axis. The Format Axis task pane opens for you to format the value axis.
- Click the Display units arrow and select Thousands.
- Click the Home tab, select the title Thousands, and then apply Dark Blue font color in the Font group. Close the Format Axis task pane. Save the workbook.



STEP 8: ADD AND FORMAT DATA LABELS

The pie chart includes a legend to identify which color represents each computer-related job; however, it does not include numerical labels to help you interpret what percentage of all computer-related jobs will be hired for each position. You want to insert and format percentage value labels.

- a. Click the Pie Chart sheet tab and click Chart Elements.
- b. Click the Data Labels arrow and select Center. Close the Chart Elements menu. You added data labels to the pie slices. The default data labels show the number of new jobs in the pie slices.
- c. Right-click one of the data labels and select Format Data Labels to open the Format Data Labels task pane.
- d. Click Label Options, click the Percentage check box to select it, and then click the Value check box to deselect it. Close the Format Data Labels task pane. Typically, pie chart data labels show percentages instead of values.
- e. Change the font size to 18 to make the data labels larger. Save the workbook.



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